# Risk data store Platform

## Architecture recommendation

Prepared by Placida Fernando, Enterprise/Solution Architect

# Problem Statement

The bank wants to build a Risk data mart by consolidating data from three source types: its existing core banking systems, the enterprise MDM (master data management) platform, and CSV file extracts received from various lines of business (LOBs).

# Architecture PoV

Assumptions

- System of records already have Active Directory federated accessibility to the landing zone
- Consumers will have role-based authentication
- Existing data store infrastructure will be used without investing for new technology during the initial phase
- DR and High availability platform will mirror the newly built data platform

# Architecture PoV and assumptions

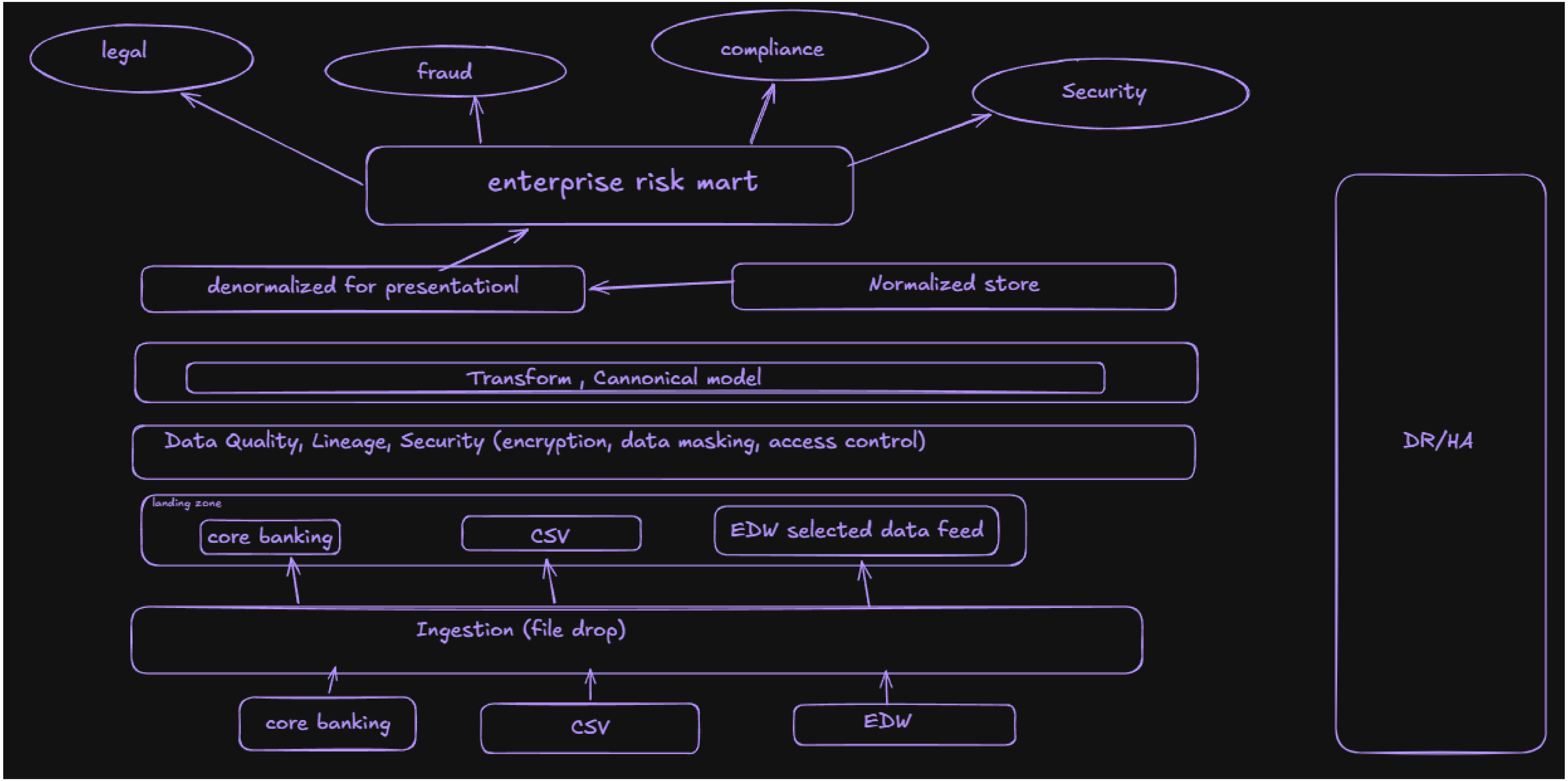
**1. "System of records already have Active Directory federated accessibility to the landing zone"** : *Source systems (core banking, MDM) are reachable from the data platform's landing zone, with service accounts and authentication federated through the bank's enterprise identity provider (e.g., Active Directory).*

**2. "Consumers will have role-based authentication"** - *Consumer access to the data mart is governed by role-based access control (RBAC), with roles mapped to existing enterprise directory groups; row/column-level restrictions apply for sensitive attributes.*

**3. "Existing data store infrastructure will be used without investing for new technology during the initial phase"** - *Phase 1 leverages the bank's existing data platform infrastructure and licensed tooling; no new technology procurement is assumed. Technology modernization is deferred to a later phase.*

**4. "DR and High availability platform will mirror the newly built data platform"** - *The new data platform inherits the bank's standard HA and DR provisions; DR environments mirror production, and the platform's RPO/RTO targets align with the bank's existing standards for risk data.*

# Integration Diagram (data flow is bottom to top)

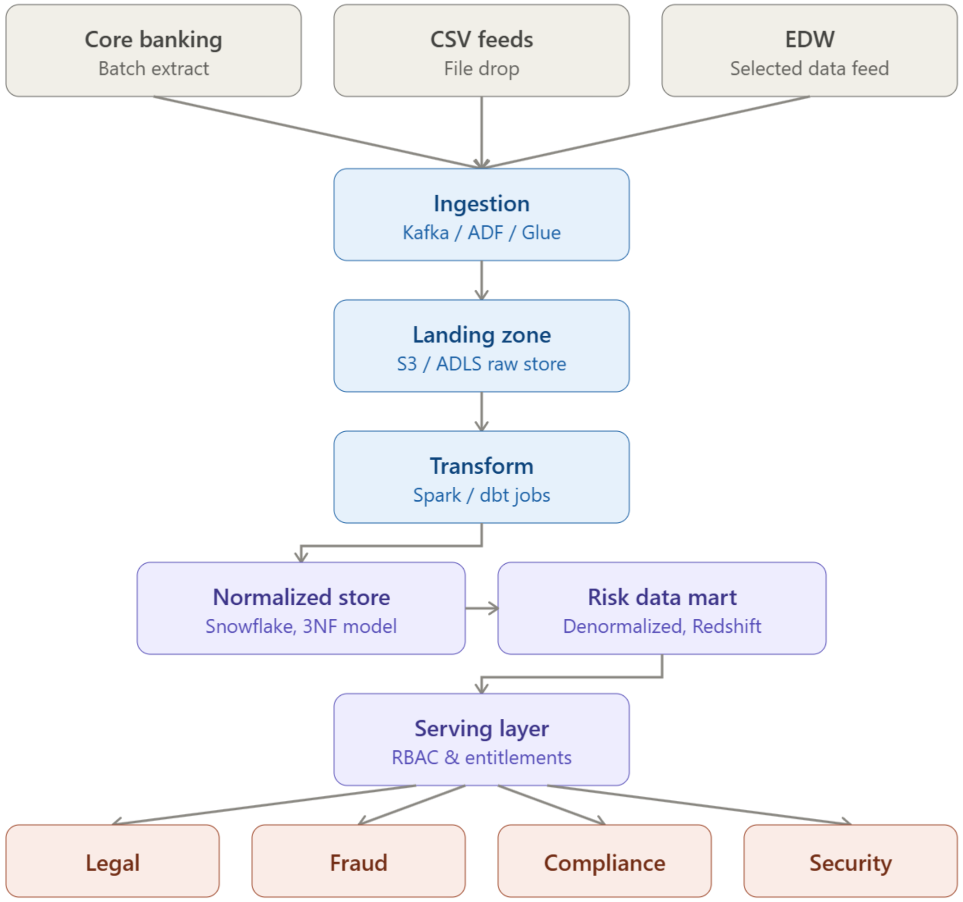



Prepared by Placida Fernando, Enterprise/Solution Architect

# Technology architecture

| Layers | Diagram symbols | What "technology architecture" adds ? |
|---|---|---|
| Sources | core banking, CSV, EDW | connection type (JDBC, SFTP, API), frequency (batch/real-time) |
| Landing zone | a box | actual storage tech (e.g., S3/ADLS/HDFS), format (Parquet/Avro), ingestion tool (Kafka, NiFi, ADF, Glue) |
| Normalized store | "new data store" | the actual database/warehouse engine (Snowflake, Redshift, Synapse, Oracle) + modeling approach (3NF) |
| Denormalized mart | "enterprise risk mart" | data mart/OLAP tech, transformation engine (dbt, Spark, stored procs) |
| Consumers | legal, fraud, compliance, security | how they access it - BI tool, API layer, direct SQL, RBAC/entitlements |

# Technology architecture




Gray = source systems · Blue = movement / processing · Purple = storage & serving · Coral = consumers

Prepared by Placida Fernando, Enterprise/Solution Architect